Korean Normative Hearing Data & ISO 7029:2017

Reference standards, peer-reviewed validation, and clinical infrastructure behind Audiso's screening and rehabilitation technology.

Prepared by Young Joon Seo, MD — CEO, Audiso Co., Ltd.
Professor of Otolaryngology, Yonsei University Wonju College of Medicine
Director, Korean Hearing Reference Data Center (since 2017)

Audiso Co., Ltd. · Yonsei University Wonju College of Medicine · 2026
For partnership and licensing inquiries: audiso.seo@gmail.com · audimall.vercel.app

Why Korean Reference Data Matters for Global Audiology

Audiso's screening AI, kiosk deployments, and digital rehabilitation modules are built on nationally designated hearing reference data — not generic thresholds copied from legacy audiometry tables. This brief summarizes the peer-reviewed evidence and clinical infrastructure that underpins our technology stack.

KEY FACTS

1,089
Pure-tone audiometry (PTA) reference standards for Korean adults, by age, sex, and frequency

72
ABR reference standards (Korean adults, age/sex/frequency stratified)

100,000
Hearing-loss-related records structured over three years (lifelog big-data platform project)

CLINICAL LEADERSHIP

Young Joon Seo, MD, PhD, is Professor & Chief of Otorhinolaryngology at Yonsei University Wonju College of Medicine, Director of the Research Institute of Hearing Enhancement, and CEO of Audiso Co., Ltd. (since August 2022). He directs Korea's national **Korean Hearing Reference Data Center** (since 2017) and serves as an **ISO TC43 WG1 member** on the international hearing standards committee.

Clinical practice includes **1,350+ ENT surgeries** (cochlear and bone-anchored implant surgery among otology procedures). Audiso operates Korea's **KOLAS-accredited audiometer calibration laboratory (ISO/IEC 17025)**.

PUBLICATION TRACK RECORD

The research team has published **50+ SCI/SCIE publications** (CV); 53 SCI(E) papers 2015–2023 per national science award dossier. Recent peer-reviewed work directly relevant to screening calibration includes:

- Yoon CY et al., *J Clin Med* 2025 — AI-based prediction of bone conduction thresholds from air-conduction audiometry
- Lee D et al., *Am J Audiol* 2026 — Normal hearing thresholds in Korean adults aged 20–79 vs. ISO 7029:2017
- Lee HS et al., *Otol Neurotol* 2026 — Comprehensive audiological outcomes of the Osia 2 System (prospective clinical study)

National Hearing Reference Data Center

Since 2017, Korea's designated hearing reference data center has produced normative datasets used for clinical comparison, device calibration, and AI model training. The center's work includes age-, sex-, and frequency-specific standards aligned with international metrology practice through **ISO TC43 WG1** participation.

DATASET SCOPE

Dataset	Count	Stratification
Pure-tone audiometry (PTA) reference standards	1,089	Age, sex, frequency (Korean adults)
ABR reference standards	72	Age, sex, frequency (Korean adults)
Structured hearing-loss records	~100,000	3-year lifelog big-data platform collection

DIGITIZATION PIPELINE

The team developed software to convert audiogram image files into structured data for reference-standard comparison — enabling large-scale validation of screening outputs against published norms rather than ad-hoc clinic baselines.

ISO ALIGNMENT

Korean age-, sex-, and frequency-specific hearing reference standards were developed with contribution to **ISO TC43 WG1**. This work provides the metrological foundation for Audiso's ISO-aligned screening stack and KOLAS-accredited calibration lab.

Source: ceo_credibility_profile.json — hearing_big_data, clinical.lab, titles

Korean Normative Thresholds vs. ISO 7029:2017

ISO 7029:2017 defines age-dependent hearing threshold levels for screening purposes. Applying ISO tables without population-specific validation can misclassify hearing loss in non-Western cohorts — particularly in Asia-Pacific markets where device OEMs and clinic networks seek locally validated norms.

PEER-REVIEWED COMPARISON (LEE D ET AL., AM J AUDIOL 2026)

"Normal Hearing Thresholds in Korean Adults Aged 20–79 Years: Establishing Reference Values and Comparison With ISO 7029:2017" establishes Korean normative hearing thresholds and compares them directly against ISO 7029:2017 reference values.

This publication underpins Audiso's regional calibration strategy:

- Screening kiosks (WithHear) use norms validated against Korean reference data, not generic ISO tables alone
- AI audiometry models are trained and validated on the same structured dataset (100,000+ records)
- Partnership discussions with global OEMs can reference published Am J Audiol 2026 evidence

CLINICAL AI VALIDATION (YOON CY ET AL., J CLIN MED 2025)

"AI-Based Prediction of Bone Conduction Thresholds Using Air Conduction Audiometry Data" demonstrates that air-conduction screening data — when calibrated against reference standards — can support clinical decision pathways without full bone-conduction testing in every case.

Publications cited from verified CEO credibility profile. Full citations available on request.

KOLAS-Accredited Calibration Laboratory

Audiso operates a **KOLAS-accredited audiometer calibration laboratory** accredited under **ISO/IEC 17025**. This is the same rigor expected by German audiology networks and NHS procurement teams evaluating device accuracy claims.

WHY CALIBRATION MATTERS FOR AI SCREENING

- AI models trained on miscalibrated hardware produce systematically biased outputs
- Reference-standard comparison requires traceable audiometer calibration
- KOLAS accreditation provides third-party verification of measurement traceability

IMPLANT & SURGICAL EVIDENCE

Beyond screening, the clinical team publishes implant outcomes data. Lee HS et al., *Otol Neurotol* 2026 — **"Comprehensive Audiological Outcomes of the Osia 2 System: A Prospective, Single-Arm Clinical Study"** — supports diligence conversations for M&A and licensing partners evaluating clinical depth.

AWARDS & RECOGNITION

- Minister of Health and Welfare Award (2025)
- Ministry of Trade, Industry and Energy Awards (2019, 2024)
- Research Excellence Award, Yonsei University

From Reference Data to Deployable Products

Audiso's four-stage clinical ecosystem translates national reference data into partnership-ready technology layers:

1. WithHear — AI Kiosk Screening

Three-minute, staff-free screening on **ISO-aligned hearing data**. Deployed in lobbies, pharmacies, and public-health sites; routes qualified leads into clinic networks.

2. Modoo Hearing — Data-Driven Matching

Matches patients with optimal hearing centers using structured audiometric data and reference-standard comparison.

3. Peacetop Triple Care — Nutrition Layer

Proprietary nutrition supporting ear, vascular, and nerve balance — extending the clinical pipeline beyond hardware.

4. MindTone T Care — SaMD Rehabilitation

Software as a Medical Device (SaMD) for tinnitus and hearing rehabilitation at home — portfolio extension where hardware-only models plateau.

The ecosystem was validated at **CES · KIMES · WCA 2026 Seoul** and is public at **AudiMall** (audimall.vercel.app) — the showroom for technology built on Korea's only KOLAS-accredited audiometer lab in this category.

Implications for Global Partners

FOR OEMS & DISTRIBUTORS

Korean normative data (1,089 PTA refs) plus ISO 7029 comparison (Am J Audiol 2026) provides evidence for Asia-Pacific market entry — a practical execution layer beyond Western-only norms.

FOR DTX & SAMD PARTNERS

AI bone-conduction prediction (J Clin Med 2025) and structured 100,000-record dataset support co-development of digital therapeutics on validated clinical infrastructure.

FOR M&A & CORPORATE DEVELOPMENT

Professor-led platform: 50+ SCI papers, 1,350+ surgeries, national reference center directorship, KOLAS lab, and four-layer product pipeline. Full 12-page IR deck available on request.

NEXT STEPS

- Download the full whitepaper PDF at audimall.vercel.app/en/whitepaper/
- Explore partnership program: audimall.vercel.app/en/partners/
- Contact: audiso.seo@gmail.com · Young Joon Seo, MD